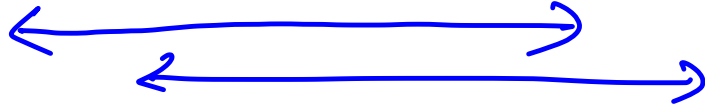
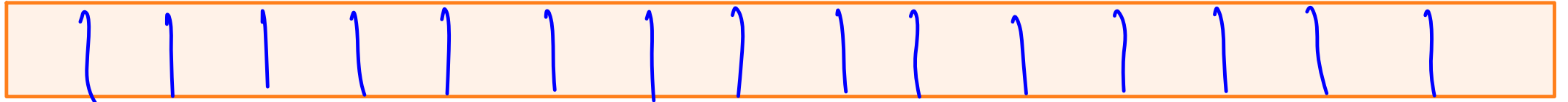




Advanced Binary Search 4

Problem 1: Maximum Average Segment

n, d



$d = 5$
 $n = 10$

$d = 5$

↳ # of subarray of length $d = \underline{\underline{n - d + 1}}$

[5, 6, 7, 8, 9, 10]

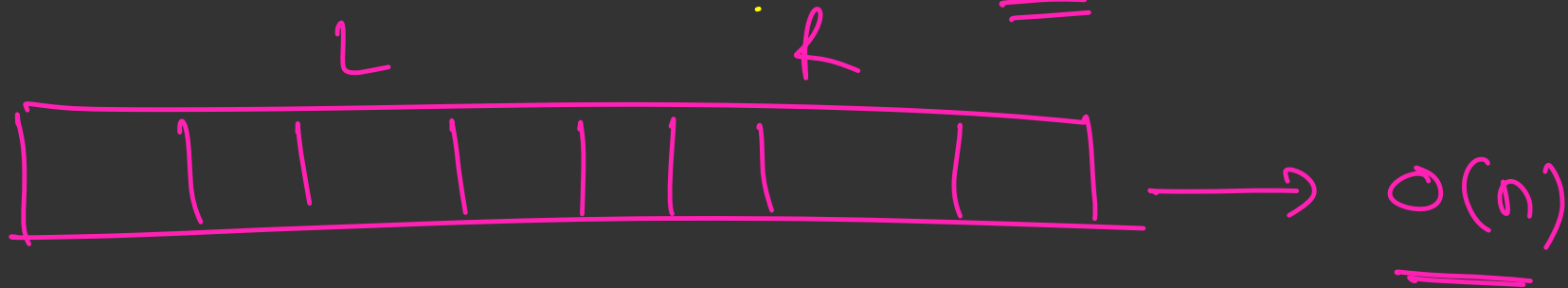
$d \rightarrow \underline{\underline{(n-d+1)}}$

checking subarray mean = $O(1)$

$\hookrightarrow$ sum using prefix sum

$O(\underline{\underline{n-d+1}})$

$\downarrow$ $O(1)$



$p[R] - p[L-1]$

$$\text{if } d = 5 \rightarrow \textcircled{n}$$

$$\begin{array}{l} O(n-5+1) \\ O(n-(5+2)+1) \\ \vdots \end{array}$$

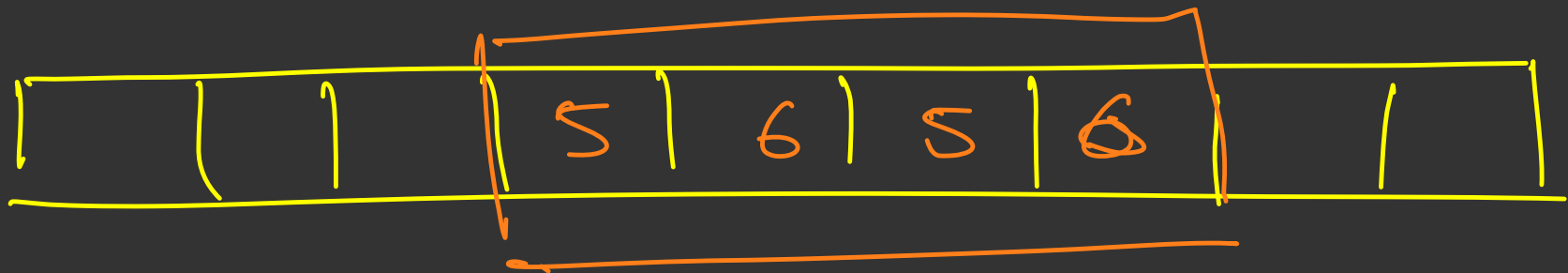
$$\boxed{d=1}$$

$$\underline{\underline{O(n^2)}}$$

We want a subarray whose mean is maximum & the size is at least d

if there exists a subarray of length $\geq d$ whose mean is $\geq X$

Can I say there will also be a subarray with mean $\geq X - \text{something}$



$$\geq \boxed{5.5}$$

$$\geq \boxed{4}$$

$$\boxed{\geq x}$$

... (7) 8 (9) (10) ... 15 (17)

T T T T ... T F F F F F F

$f(x) \rightarrow T$ if there exists a subarray with
 mean $\geq x$

you are considering a mean as x

$f(x)$ whether there exists a subarray
whose mean $\geq x$ $f(x)$

$$[a_i, a_{i+1}, \dots, a_{i+k-1}]$$

to a subarray of length k

$$k \geq d$$

$$\frac{a_i + a_{i+1} + a_{i+2} + \dots + a_{i+k-1}}{k} \geq x$$

$$a_i + a_{i+1} + a_{i+2} + \dots + a_{i+k-1} \geq kx$$

$$[a_{i-x}] + a_{i+1} + a_{i+2} \dots a_{i+k-1} \geq (k-1)x$$

$$[a_{i-x}] + [a_{i+1-x}] \dots [a_{i+k-1-x}] \geq 0$$

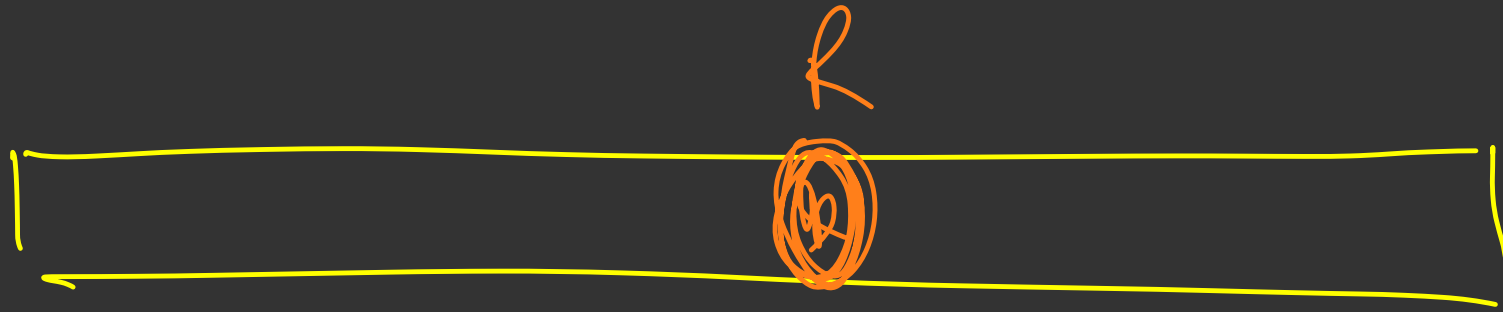
$f(x) \rightarrow T$ if there exists a subarray of
length at least d with
mean $\geq x$

$$A \rightarrow [a_1 \quad a_2 \quad \dots \quad a_n]$$

$$\downarrow$$

$$B \rightarrow [a_1 - x \quad a_2 - x \quad \dots \quad a_n - x]$$

find out the maximum subarray sum
s.t length of subarray is atleast d.

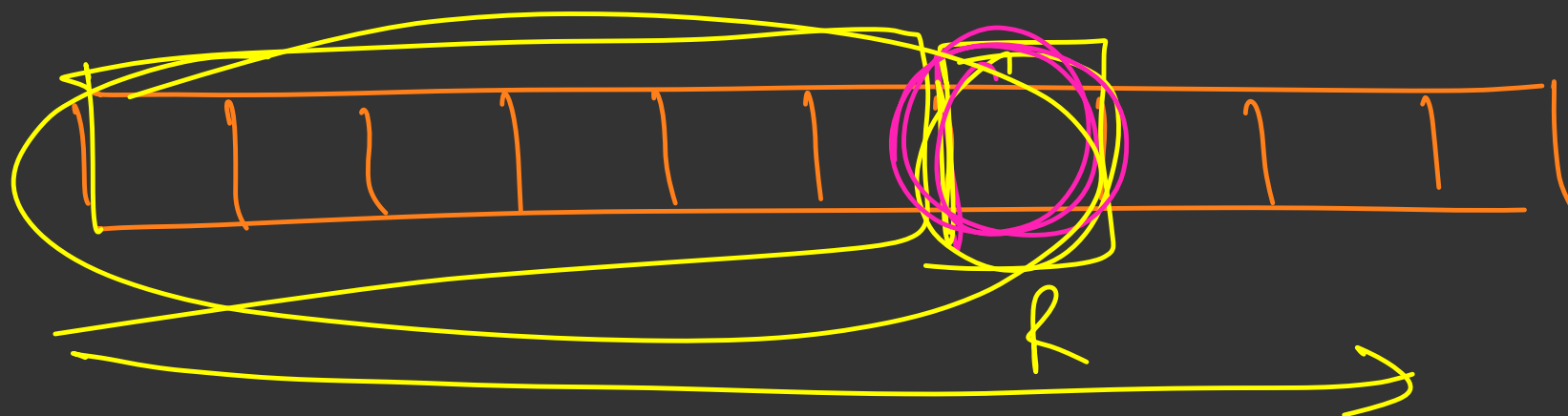


$$\text{Subarray sum } [L \text{ to } R] \rightarrow P[R] - P[L-1]$$

The formula $P[R] - P[L-1]$ is shown. The term $P[R]$ is circled in red. The entire expression is enclosed in a red box. An arrow points from the box to the text "max".

$$\boxed{P[R] - P[L-1]} \rightarrow \text{max}$$

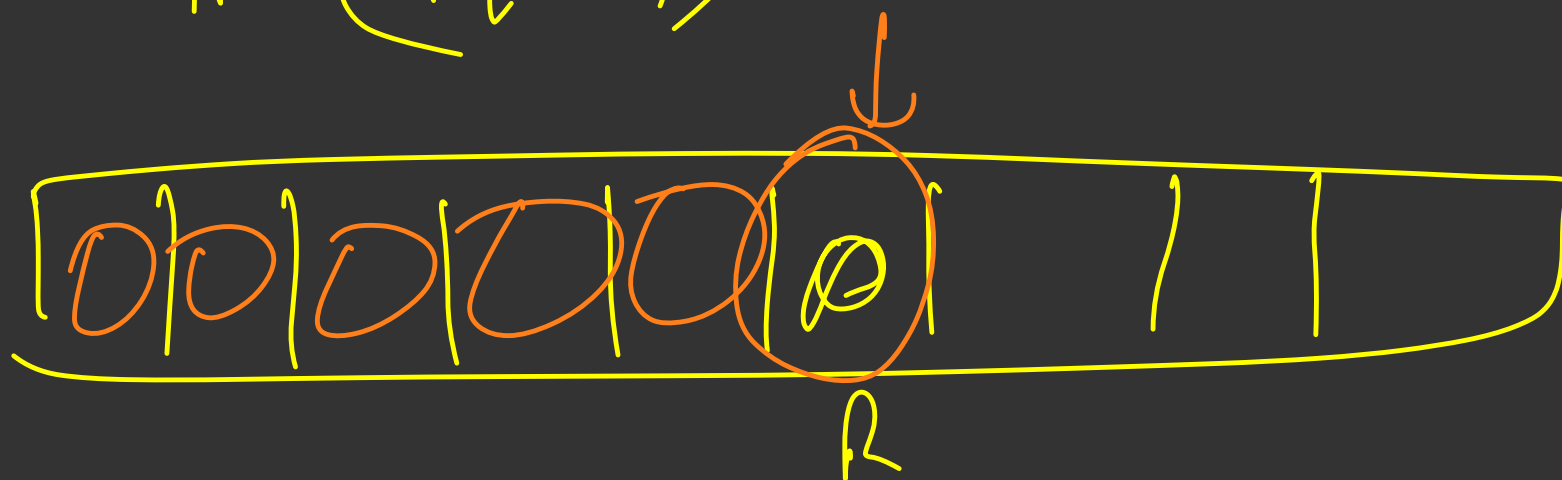
if we want max subarray sum s.t subarray
ends on R $\rightarrow$ we want to minimize $P[L-1]$

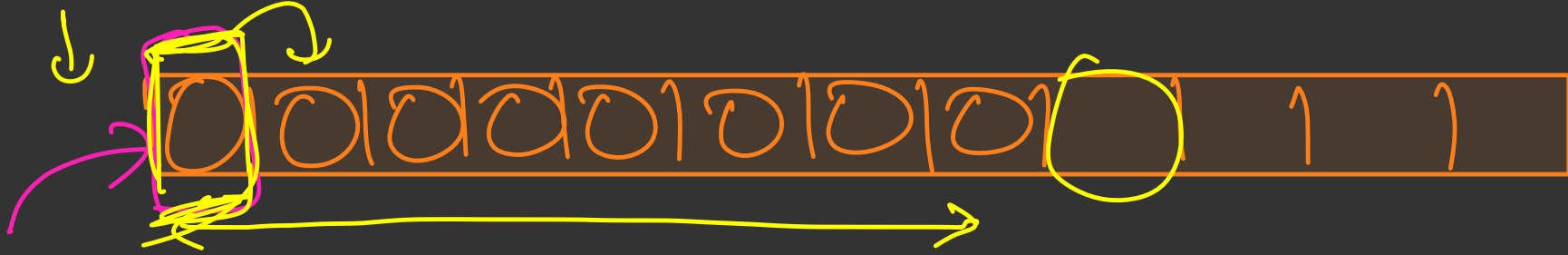


$$(P[R] - P[L-1]) \rightarrow \text{max}$$

$$\downarrow$$

$$\text{if } (P[L-1]) \rightarrow \text{min}$$





$$(P[R] - P[L-1]) \rightarrow \max$$

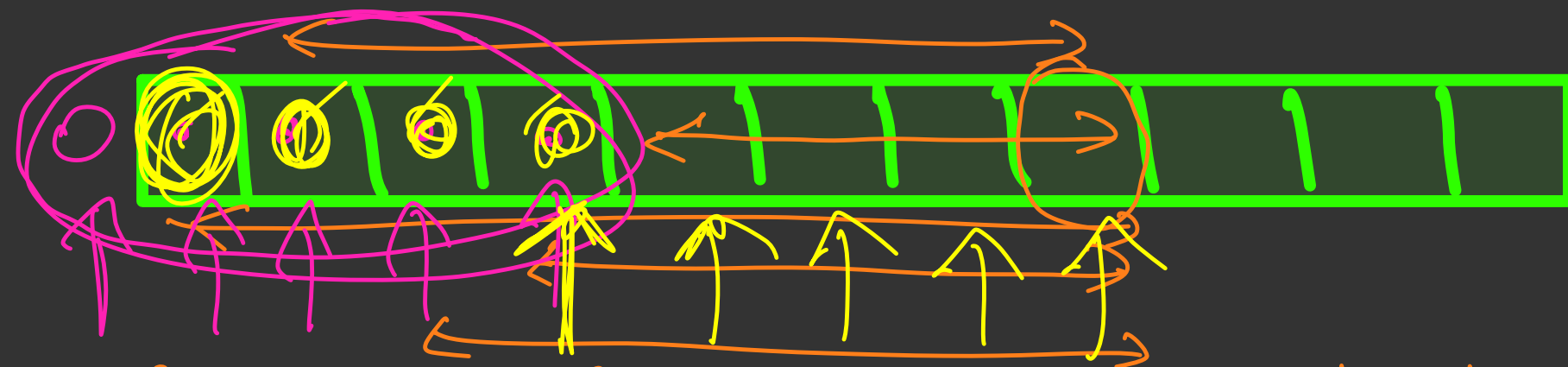
$$P[R] - (P[L-1] \rightarrow \min)$$

$$\underline{\minsofar} = 0$$

only those subarray whose length is atleast d.



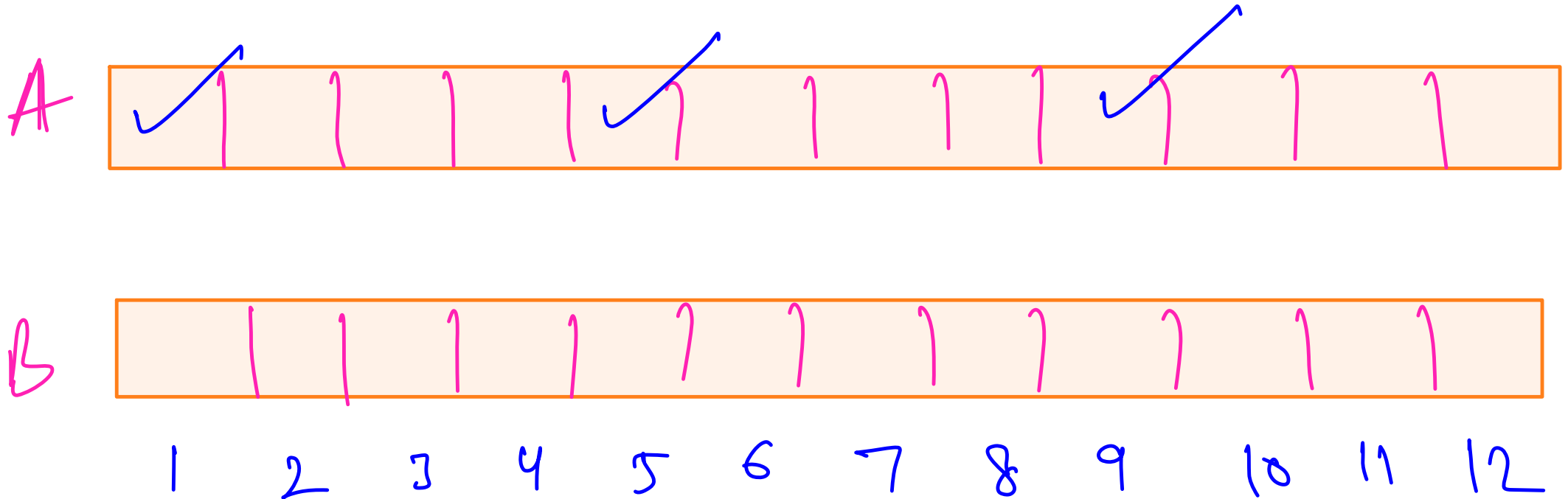
$$\min_{\text{Info}} = 0$$



$$d = 4$$

~~$p[r] - p[l-1]$~~ — minsofen

Problem 2: Pair Selection



$$\left[\frac{a_1 + a_5 + a_9}{b_1 + b_5 + b_9} \right]$$

If there exist a choice with ratio
 $\geq X$

there also exist a choice with
ratio $\geq X$ - something

$f(n)$ $\leftarrow$ $\begin{cases} \text{True} & \text{if there exist a ratio} \\ & \geq X \\ \text{False} & \text{else} \end{cases}$

TTTTTTTTT T fffffffffff

$f(x)$ $\xrightarrow{f}$ 'ratio' $\geq x$



10.5



$$\frac{a_{i_1} + a_{i_2} + a_{i_3} \dots a_{i_k}}{\delta_{i_1} + \delta_{i_2} + \delta_{i_3} \dots \delta_{i_k}} \geq x$$

$$a_{i_1} + a_{i_2} - \dots - a_{i_k} \geq n [\delta_{i_1} + \delta_{i_2} - \dots - \delta_{i_k}]$$

$$[a_{i_1} - \delta_{i_1}n] + [a_{i_2} - \delta_{i_2}n]$$

$$- \dots - [a_{i_k} - \delta_{i_k}n] \geq 0$$

(n)

A

B

y

→

c

$$[a_1 - \delta_1 n, a_2 - \delta_2 n$$

$$- \dots - a_k - \delta_k n]$$

Problem 3 [HW]: Average & Median